

Zhixu Duan

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EDUCATION

- **University of Electronic Science and Technology of China** Chengdu, China
Bachelor of Mechanical Design, Manufacturing and Automation, School of Mechanical and Electrical Engineering Sep. 2023 - Present

GPA: 3.97/4.00; Weighted Average: 90.7/100; Major Ranking: 1/83; CET-6: 512

RESEARCH EXPERIENCE

- **Shanghai Jiao Tong University, School of Artificial Intelligence** Shanghai, China
Research Assistant at EPICLab; advised by Prof. Linfeng Zhang; efficient and precision intelligent computing Nov. 2025 - Present
- **University of Electronic Science and Technology of China** Chengdu, China
Research Assistant at ReliaLab / Center for System Reliability and Safety; advised by Prof. Zuoyi Chen and Prof. Hong-Zhong Huang Jan. 2024 - Present

RESEARCH INTERESTS

My research interests focus on AI, robotics, automation and mechanical engineering related topics.

PUBLICATIONS AND PROJECTS

(* Equal contribution / first-student author; † corresponding author)

- [1] ***Decoupling Intrinsic Fault Features from Domain Variations via Domain-Attribute Fusion for Unseen-Domain Fault Diagnosis***
Accepted by **Advanced Engineering Informatics** (SCI/CAS Q1, IF: 9.9); patent filed
Zhixu Duan, et al.
- [2] ***Pseudo-fault Data Enhanced Relation Network for Fault Detection and Localization in Train Transmission Systems***
Accepted by **Engineering Applications of Artificial Intelligence** (SCI/CAS Q1, IF: 7.5)
Zhixu Duan, Ruoxi Liu, Zuoyi Chen, Hong-Zhong Huang†
- [3] ***Unified Health Domain Relation Learning for Train Transmission Systems Fault Detection under Complex Operating Conditions***
Accepted by **Structural Health Monitoring** (SCI/CAS Q2, IF: 5.7); patent filed
Zuoyi Chen*, **Zhixu Duan***, Hong-Zhong Huang†
- [4] ***Parallel Relation Network for Intelligent Fault Detection and Localization of Train Transmission Systems with Zero-fault Sample***
IEEE PHM 2024 Oral Presentation
Zhixu Duan, Ruoxi Liu, Zuoyi Chen, Hong-Zhong Huang†
- [5] ***Reinforcing Cross-Domain Few-Shot Fault Diagnosis of Train Transmission Systems via Reducing Intra-Class and Maximizing Inter-Class Variations***
Accepted by **Advanced Engineering Informatics** (SCI/CAS Q1)
Ruoxi Liu, **Zhixu Duan**, Zuoyi Chen, Hong-Zhong Huang†
- [6] ***Collaborative Teacher-Student Learning: Simulated Domain Attacks for Class-Intrinsic Feature Learning in Multi-Domain Generalized Fault Diagnosis***
Submitted to **IEEE Transactions on Industrial Informatics** (SCI/CAS Q1, IF: 9.9); patent filed
Zhixu Duan, et al.
- [7] ***Open-Set Fault Diagnosis Using CLIP with Forward-Reverse Reasoning***
Minor revision at **Computers in Industry** (SCI/CAS Q1)
Zuoyi Chen*, **Zhixu Duan***, Hong-Zhong Huang†
- [8] ***IndustryCode: A Benchmark for Industry Code Generation***
Technical Report / Benchmark Project
Contributor

HONORS AND AWARDS

- National Scholarship, 2025; National Scholarship, 2024; two-time recipient, top 1.8–1.9%.
- First-Class Outstanding Student Scholarship, UESTC, top 10%.
- “Mechanical and Electrical Elite”, selected as one of 10 students across undergraduates, master’s, and PhD students in SMEE.
- Principal leader of a National Innovation and Entrepreneurship Program for College Students; completed with excellent conclusion.

LEADERSHIP, OUTREACH, AND SKILLS

- Founding Core Member and Vice President, UESTC Interdisciplinary Association; independently built and maintained the official website; built an advisory board with 5 professors including Prof. Dezhong Yao and Prof. Pedro Antonio Valdes-Sosa. Feb. 2025 - Present
- Co-hosted the UESTC-IA × DP Technology Uni-Lab Developers’ Offline Workshop and led a delegation to Yibin; interviewed by SMEE Idol in “Take Each Step Steadily, and the Distance Will Unfold”. 2025
- Student Mentor for SMEE 2025 freshmen. Sep. 2025 - Present
- Skills: Python, PyTorch, Jupyter, C, MATLAB, Visio, Origin, PowerPoint, Git, Linux, SolidWorks.